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Date: 09/18/2026

LABORATORY ACTIVITY #4

ASCII Symbol Visualizer

Design and implement a Python program that prompts the user to input a **sentence** and a **display symbol** (a single character). The program will analyze the sentence character by character, calculate a unique numeric value for each character based on its **ASCII value**, and print a horizontal bar of repeating symbols matching that calculated value.

Requirements:

1. **Inputs:** A text string (sentence) and a single-character string (symbol).
2. **Modulo Formula:** Convert each character to its ASCII value and apply **modulo 15** ($\text{ascii_value} \% 15$) to determine the symbol count.
3. **Zero Boundary Handling:** If the modulo operation results in 0, the count must automatically scale up to **15** to ensure every character displays at least some symbols.

Sample Output

```
Enter a sentence: hello
Enter the symbol or character to display: *

--- Output ---
'h' (Count: 14) -> *****
'e' (Count: 11) -> *****
'l' (Count: 3) -> ***
'l' (Count: 3) -> ***
'o' (Count: 6) -> *****
```

Code and Output:

```
ASCII.py  ▢ X
1      print("ITEC 111")
2      print("ASCII SYMBOL VISUALIZER\n")
3
4      sentence = input("Enter a sentence: ")
5      symbol = input("Enter the symbol or character to display: ")
6
7
8      symbol = symbol[0]
9
10     print("\n--- Output ---")
11
12     for char in sentence:
13         ascii_value = ord(char)
14         count = ascii_value % 15
15
16         if count == 0:
17             count = 15
18
19         bar = symbol * count
20
21         print(f"'{char}' (Count: {count}) -> {bar}")
22
```

```
PS C:\Users\DELL\Downloads> python ASCII.py
ITEC 111
ASCII SYMBOL VISUALIZER

Enter a sentence: hello
Enter the symbol or character to display: *

--- Output ---
'h' (Count: 14) -> *****
'e' (Count: 11) -> *****
'l' (Count: 3) -> ***
'l' (Count: 3) -> ***
'o' (Count: 6) -> *****
PS C:\Users\DELL\Downloads>
```